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Total no. of Pages: 1 Roll no. ....

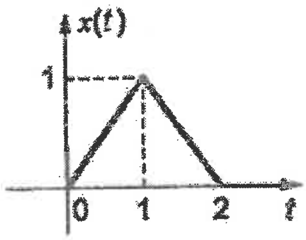
Degree: B-Tech. Semester 3^{rd}

MID-SEMESTER EXAMINATION, June, 2026

Course Title: Signals and Systems
Course Code: Sem: ECECC05

Duration: 1:30 Hours Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Define a continuous-time signal and a discrete-time signal with one example each.	**1.5**	**C01**
1b	Define the unit impulse signal and state any two properties of the impulse function.	**1.5**	**C01**
2a	Determine whether or not each of the following systems are time variant with input x(t) or x[n] and output y(t) or y[n]. (i) $y[n] = x[n] - x[n-1]$ (ii) $y[n] = x[t-1] - x(1-t)$	**1.5**	**C01**
2b	Draw the even and odd components of the waveform of a signal x(t) shown in figure. 	**1.5**	**C01**
3a	Determine the output of an LTI system if $x(t) = \delta(t)$ and $h(t) = e^{-t}u(t)$	**1.5**	**C04**
3b	Define a Linear System. State the principle of superposition.	**1.5**	**C01**
4a	State and prove distributive property of continuous time convolution	**1.5**	**C02**
4b	The unit impulse response of a system is given as $c(t) = -4e^{-t} + 6e^{-2t}$. Find the step response of the system for $t \geq 0$.	**1.5**	**C04**
5a	What is meant by orthogonality of signals? Give one example.	**1.5**	**CO2**
5b	For the input $x(t) = \delta(t-1) + \delta(t+1)$ and impulse response $h(t) = u(t+1) - u(t-2)$, draw the waveform corresponding to output $y(t)$.	**1.5**	**C02**